



TaxJar

Salesforce Commerce Cloud Installation Guide and Documentation

Version 21.1.1

Table of Contents

1. Overview

- 1.1. What is TaxJar?
- 1.2. Use Cases
- 1.3. Limitations
- 1.4. Compatibility
- 1.5. Fallback and Recovery

2. Getting Started

- 2.1. TaxJar Account Setup
- 2.2. Input Account Information
- 2.3. Retrieve TaxJar API Token

3. SFRA Cartridge Installation

- 3.1. Upload Cartridge using Eclipse
- 3.2. Update Cartridge Path

4. Site Genesis Cartridge Installation

- 4.1. Upload Cartridge using Eclipse
- 4.2. Add Custom Code
- 4.3. Update Cartridge Path

5. Configuration

- 5.1. Import MetaData

- 5.2. Configure Settings
- 5.3. Configure TaxJar Sync Job

1. Overview

1.1. What is TaxJar?

TaxJar makes sales tax filing easier for more than 20,000 online sellers and merchants across the world. TaxJar provides online solutions to sales tax compliance including tax calculation, nexus determination, and sales tax reporting and filing.

1.2. Use Cases

The use cases of this cartridge are two fold:

- Provide sales tax calculation at the point of sale (in cart/checkout) for storefronts. Cart data is sent to the TaxJar API and the returned tax rates are then applied to the cart.
- Sync relevant sales tax data from orders to TaxJar. This allows TaxJar to generate sales tax reports by location and automatically file on behalf of businesses (must be enrolled in our [AutoFile](#) service). See limitations section for more details about constraints regarding this feature in SFCC.

1.3. Limitations

Tax calculations are only available for the US, Canada and VAT tax. Full list of supported countries can be found [here](#). All other areas will fall back to default Commerce Cloud tax calculation functionality.

This cartridge does not currently support in-store pickup as a shipping method.

Due to the way most ERP (Enterprise Resource Planning) and ORM (Order Management System) systems integrate with SFCC there is often not enough data

in SFCC to sync to TaxJar for accurate reporting and filing within TaxJar. Due to this limitation it is recommended to not sync transaction data directly from SFCC (through the TaxJar Sync Job) and instead send transaction data directly from the ERP or OMS.

1.4. Compatibility

The TaxJar Commerce Cloud cartridges have been tested against and confirmed fully compatible with with following:

Compatibility Modes:

- 21.2

SFRA Versions:

- 4.2
- 4.3
- 4.4
- 4.4
- 5.0
- 5.1
- 5.2
- 5.3

Site Genesis Versions:

- 104.1.3
- 105.2.0

1.5. Failover and Recovery

TaxJar supports an SLA of 99.99% uptime and the status of TaxJar services can be found here: <https://status.taxjar.com/>

This cartridge relies on the TaxJar REST API in order to provide real time tax calculations. In the event of downtime the cartridge will fall back to the default tax

calculation within Commerce Cloud. This means orders will be able to be placed without disruption.

2. Getting Started

2.1. TaxJar Account Setup

In order to use the features included in the TaxJar SFRA cartridge a TaxJar account must first be set up. If you already have a TaxJar account skip down to step 2.2 to view the steps to retrieve your TaxJar API token. In order to register for a new account, visit https://app.taxjar.com/sign_up and follow the registration steps. TaxJar provides a free, no risk 30 day trial to allow you to evaluate our platform!

2.2. Input Account Information

In order to accurately calculate tax at the point of sale, TaxJar requires some information regarding your business. Visit <https://app.taxjar.com/account> to fill out your business profile. TaxJar will only calculate tax for states with [nexus](#). Configure your nexus states on the [state setting page](#).

2.3. Retrieve TaxJar API Token

In order to retrieve your TaxJar API Token, visit <https://app.taxjar.com/account> and click on “SmartCalcs API” in the menu on the left. If you have already generated a token it will be displayed here. Otherwise click the “Generate API Token” button. Your API Token will appear within a few seconds.


[Get a Free AutoFile!](#)
[Dashboard](#)
[Transactions](#)
[Filing History](#)
[Account](#)
[Sign Out](#)

During your trial, TaxJar imports the last 2 months of transactions from **Sep 1, 2017 - Nov 17, 2017**
[Import Your Data](#)

Account

Business Profile
 Notification Preferences
 State Settings
 Linked Accounts
 AutoFile Enrollments
SmartCalcs API
 Password
 Plans & Billing

SmartCalcs API Token

Generate API Token

Sellers: Use this API Token with one of our cart extensions or plugins (Magento or WooCommerce).

Developers: Use the API in your marketplace or ecommerce site.

[View our API documentation](#) for more information.

Try our API clients: Ruby : PHP : Node

SmartCalcs Usage

[SmartCalcs Dashboard](#) →

Stripe Tax Settings

Generate My Stripe URL

Enter the above URL under your Stripe Taxes Settings. ⓘ

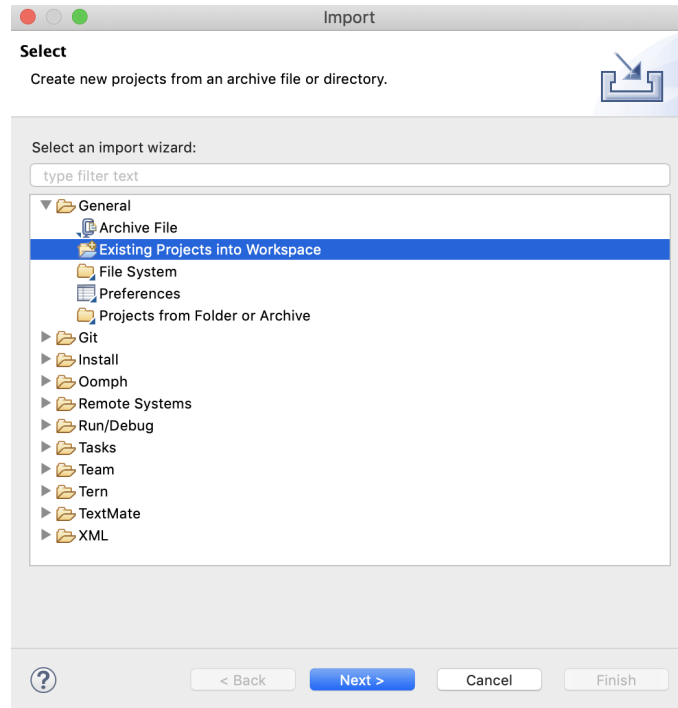
3. SFRA Cartridge Installation

3.1 Upload Cartridge Using Eclipse

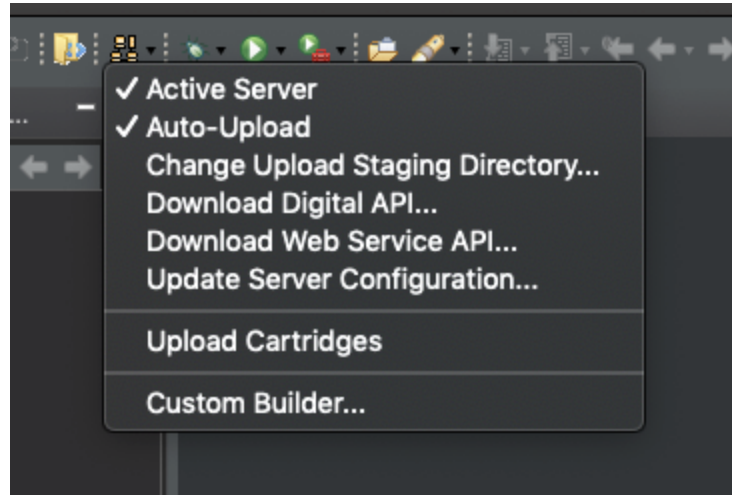
These instructions assume you have the Eclipse IDE installed and UX Studio installed. For more information on this process see the SFCC documentation at: https://documentation.b2c.commercecloud.salesforce.com/DOC1/index.jsp?topic=%2Fcom.demandware.dochelp%2FSiteDevelopment%2FGettingStartedforDevelopers.html&cp=0_1_6

Cartridge Installation Steps:

- Open UX Studio in Eclipse
- Select File > Import > General > Existing Projects into Workspace and then click Next



- Select the root directory containing the “int_taxjar” and “int_taxjar_sfra” cartridges
- Make sure that both the “int_taxjar” and “int_taxjar_sfra” projects are selected
- Click Finish
- When prompted to attach cartridges to a Digital Server, click Yes
- Select the cartridges you would like to attach (“int_taxjar” and “int_taxjar_sfra”) and click OK
- If prompted to update your server password, ensure your server host and username are correct, enter your password and click OK
- Click on the Digital Server button and then select Upload Cartridges



- If prompted to Delete Server Cartridges, ensure that no cartridges are selected and click OK

3.2 Update Cartridge Path

In Business Manager, go to Administration > Manage Sites and then select the site you would like to add the cartridge to. Navigate to the settings tab. Add “int_taxjar” and “int_taxjar_sfra” to the cartridge path, ensuring that they appear before the “app_storefront_base” base cartridge.

Cartridges:	int_taxjar_SFRA:taxjar_core:app_storefront_base
Effective Cartridge Path:	int_taxjar_SFRA:taxjar_core:app_storefront_base:plugin_apple_pay:plugin_facebook:plugin_pinterest_commerce:plugin_web_payments:bc_content:core

In order to sync transactions to TaxJar for reporting, you also need to add the “int_taxjar” cartridge to the business manager cartridge path. To do this navigate to Administration > Manage Sites and then click on the link to “Manage the business manager site”. You will now need to append “:int_taxjar” to the end of the cartridges fields as shown below. Then click the apply button.

Cartridges:	bm_custom_plugin:taxjar_core
Effective Cartridge Path:	app_business_manager:plugin_apple_pay:plugin_facebook:plugin_pinterest_commerce:plugin_web_payments:bc_impe:bc_search:bc_analytics:bc_transaction:bc_catalog:bc_content:core:bm_custom_plugin:taxjar_core:customization_defaults
Apply Reset	

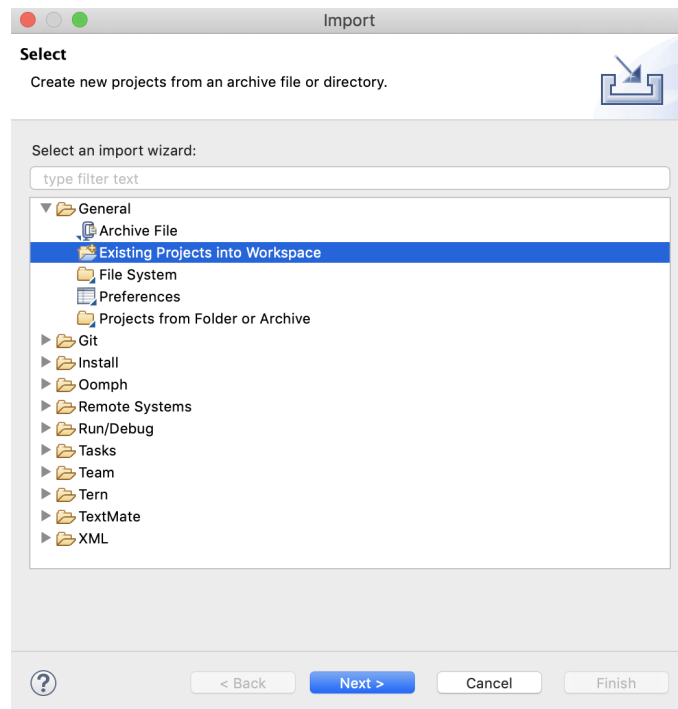
4. Site Genesis Cartridge Installation

4.1 Upload Cartridge Using Eclipse

These instructions assume you have the Eclipse IDE installed and UX Studio installed. For more information on this process see the SFCC documentation at: https://documentation.b2c.commercecloud.salesforce.com/DOC1/index.jsp?topic=%2Fcom.demandware.dochelp%2FSiteDevelopment%2FGettingStartedforDevelopers.html&cp=0_1_6

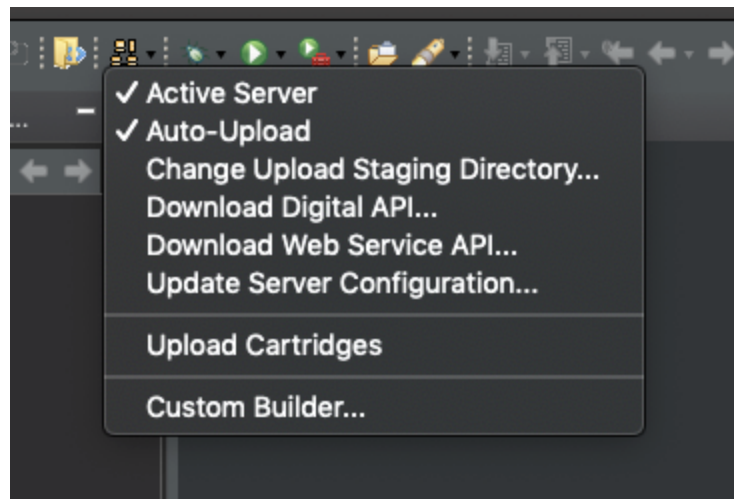
Cartridge Installation Steps:

- Open UX Studio in Eclipse
- Select File > Import > General > Existing Projects into Workspace and then click Next



- Select the root directory containing the “int_taxjar” and “int_taxjar_sg” cartridges

- Make sure that both the “int_taxjar” and “int_taxjar_sg” projects are selected
- Click Finish
- When prompted to attach cartridges to a Digital Server, click Yes
- Select the cartridges you would like to attach (“int_taxjar” and “int_taxjar_sg”) and click OK
- If prompted to update your server password, ensure your server host and username are correct, enter your password and click OK
- Click on the Digital Server button and then select Upload Cartridges



- If prompted to Delete Server Cartridges, ensure that no cartridges are selected and click OK

4.2 Add Custom Code

In the calculate.js file

(sitegenesis_storefront_core/cartridge/scripts/cart/calculate.js) you will need to replace one line of code. The line of code is found inside of the calculate function as is as follows:

```
calculateTax( basket );
```

Replace that line with the following:

```
if (!require('*/cartridge/scripts/taxJarSGJC').calculateTax(basket))
calculateTax(basket);
```

Which should give you code similar to the following:

```
77 // =====
78 //          CALCULATE TAX          //
79 // =====
80
81 if ( ! require('*/cartridge/scripts/taxJarSGJC').calculateTax( basket ) ) calculateTax( basket );
82
```

Note: Line numbers in screenshot may not exactly correspond with the line numbers in your code.

4.3 Update Cartridge Path

In Business Manager, go to Administration > Manage Sites and then select the site you would like to add the cartridge to. Navigate to the settings tab. Add “int_taxjar” and “int_taxjar_sg” to the cartridge path, ensuring that they appear after the app_storefront base cartridges.

Cartridges:	app_storefront_controllers:app_storefront_core:taxjar_SGJC:taxjar_core
Effective Cartridge Path:	app_storefront_controllers:app_storefront_core:taxjar_SGJC:taxjar_core:plugin_apple_pay:plugin_facebook:plugin_pinterest_commerce:plugin_web_payments:bc_content:core

In order to sync transactions to TaxJar for reporting, you also need to add the “int_taxjar” cartridge to the business manager cartridge path. To do this navigate to Administration > Manage Sites and then click on the link to “Manage the business manager site”. You will now need to append “:int_taxjar” to the end of the cartridges fields as shown below. Then click the apply button.

Cartridges:	bm_custom_plugin:taxjar_core
Effective Cartridge Path:	app_business_manager:plugin_apple_pay:plugin_facebook:plugin_pinterest_commerce:plugin_web_payments:bc_impex:bc_search:bc_analytics:bc_transaction:bc_catalog:bc_content:core:bm_custom_plugin:taxjar_core:customization_defaults

Apply Reset

5. Cartridge Configuration

5.1 Import Metadata

5.1.1 Import Metadata using Site Import & Export

All the metadata necessary for using the TaxJar integration is contained in the `taxjar_metadata.zip` file in the metadata directory. To upload this metadata:

1. Log into your SFCC Business Manager
2. Navigate to Administration > Site Development > Site Import & Export
3. Under the Import header, click the “Choose File” button
4. Select the `taxjar_metadata.zip` file in the metadata directory
5. Click the “Upload” button
6. Under the “Import” header select the `taxjar_metadata.zip` file that was just uploaded
7. Click the “Import” button
8. Click “OK”
9. The necessary metadata will now be imported

Note: Importing the metadata using the site import and export tool also imports a default job used for syncing transactions to TaxJar. If only the tax calculation portion of the integration is going to be used it is recommended to import the `services.xml` and `system-objecttype-extensions.xml` files individually. They are located in the metadata directory.

5.2 Configure Settings

5.2.1 Configure TaxJar Custom Preferences

The custom preferences imported in section 3.3.1 must be configured prior to using the TaxJar cartridge. In order to configure these settings follow these steps:

1. Log into Business Manager.
2. Select your desired site at the top left of the page.
3. Navigate to Merchant Tools > Site Preferences > Custom Preferences.
4. Click on the “TaxJar” custom preference.
5. Fill in the available preferences with the information for your business.

TaxJar Settings
Cancel
Apply to Other Sites
Save

Instance Type
Sandbox

Search by IDs...

1-6 of 6

Name	Value	Default Value	
Enable TaxJar	Yes	Yes	Edit Across Sites
Ship From Country Code			Edit Across Sites
	Two-letter ISO country code of the country where orders are shipped from.		
Ship From State Code			Edit Across Sites
	Two-letter ISO state code where orders are shipped from.		
Ship From Postal Code			Edit Across Sites
	Postal code where orders are shipped from (5-Digit ZIP or ZIP+4).		
Ship From City			Edit Across Sites
	City where orders are shipped from.		
Ship From Address			Edit Across Sites
	Street address where orders are shipped from.		

The following preferences are configurable in the cartridge:

- **Enable TaxJar** - Turn on or off TaxJar tax calculation. Enabled by default.
- **Ship From Country Code** - Two-letter ISO country code of the country where your orders ship from.
- **Ship From State Code** - Two-letter ISO state code where your orders ship from.
- **Ship From Postal Code** - Postal code where your orders ship from (5-Digit ZIP or ZIP+4).
- **Ship From City** - City where your orders ship from.
- **Ship From Address** - Street address where your orders ship from.

5.2.2 Configure Service

In order to configure the TaxJar REST API service follow these steps:

1. Log into Business Manager.
2. Navigate to Administration > Operations > Services.
3. You'll notice that the taxjar.http.api service was already created (in section 3.3.2 of this document).
4. Click on the "Credentials" tab.
5. Click on taxjar.api.token credential.
6. Enter your TaxJar API Token into the password field. The TaxJar API Token can be found using instructions in section 2.3 of this document. This is the only field that needs to be changed in the credential.
7. Click the "Apply" button.

Note: No changes need to be made to the service configuration except for the credential password in the steps above.

5.2.3 Configure Default Commerce Cloud Tax Rates

The TaxJar cartridge only calculates tax for states where your business has nexus. You can set your nexus states in the TaxJar app:

<https://app.taxjar.com/account#states>.

If you do not have nexus set for a state, or for areas outside of the US, the TaxJar cartridge reverts back to default SFCC tax calculation. This allows you to continue calculating tax beyond areas that are currently supported by TaxJar.

The TaxJar cartridge also reverts back to default SFCC tax calculation in the event that the TaxJar API is down. This allows you to configure tax rates inside of SFCC that can serve as a backup to the TaxJar tax calculation. You want to ensure that

you only add backup rates for areas in which you have nexus. Otherwise, you may collect extra tax.

In order to avoid collection of tax in areas outside of your nexus regions, we recommend adding a base US tax rate of 0%. Follow these steps in order to set up this base rate:

1. Navigate to Merchant Tools > Ordering > Taxation.
2. Click on the Tax Jurisdictions Tab.
3. If you do not already have a tax jurisdiction setup for the US, click “New” and then give it a name and code.
4. If you do not already have a US address mapping setup for this jurisdiction, click “New” under the “Address Mapping” section. Enter “US” in the Country field and click Apply.
5. Click the “Back to List” button or navigate back to Merchant Tools > Ordering > Taxation.
6. Click the “Tax Rates” tab.
7. Enter a rate of 0 for tax class in the US tax jurisdiction you set up in step 3 and click the “Apply” button.

[Merchant Tools](#) > [Ordering](#) > Taxation Settings - Tax Rates

General Tax Classes Tax Jurisdictions **Tax Rates**

Tax Rates

On this page you can define the tax rates for any combination of tax class and tax jurisdiction. Click **Apply** to save your settings or **Reset** to revert to your current settings.

	Clothing	Exempt	Standard
US	0.00000	0.00000	0.00000

Showing 1 - 1 of 1 items

You will want to ensure that there is no other rate setup for the US other than 0% rate in the steps above. Having any existing non 0 US rates can lead to unnecessary tax calculation.

5.2.4 Setup Tax Classes

There are many scenarios where jurisdictions tax items differently based on the type of item sold. In order to support these scenarios tax classes need to be set up within SFCC. The list of product categories supported by TaxJar can be found here: <https://developers.taxjar.com/api/reference/#get-list-tax-categories>

For each product category you sell follow these steps to create a tax class:

1. In Business Manager select correct site
2. Navigate to Merchant Tools > Ordering > Taxation
3. Select the Tax Classes tab
4. Click the “New” button
5. In the ID field enter the “Code” value obtained from the TaxJar categories reference for the relevant product category
6. In the name field enter category name (this does not need to match the TaxJar category name exactly)
7. Click “Apply”
8. Navigate to the Tax Rates tab.
9. For the default tax rate, enter a 0 rate for each tax class you created.

After creating the tax classes, they can be applied to your products through the standard SFCC methods.

5.2.5 Create Customer Exemptions

In some scenarios, jurisdictions will not tax individual purchasers who meet certain criteria. In order to exempt these customers from having sales tax applied on their orders a customer exemption will need to be added to the customer’s profile. To do this:

1. In Business Manager select the site for the customer that needs an exemption

2. Navigate to Merchant Tools > Customers > Customers
3. Search for and select the customer
4. Under the “TaxJar Tax Exemptions” section select both the exemption type and exemption states for the customer. If an exemption type is selected but no exemption states, it will default to being exempt in all states.

Note: The exemption states’ field is a multi select. If on a PC hold CTRL to select multiple fields (on a Mac hold CMD).

5.3 Configure TaxJar Sync Job

In order to sync transaction data to TaxJar for reporting and filing, you must create a job in your business manager. This allows you to configure how often you would like to sync data as well as easily sync historical data. While importing the metadata (Section 5.1) a default job configuration was created. Follow the steps below to configure the job:

5.3.1 Configure Job

1. Navigate to Administration > Operations > Jobs
2. Click on the “TaxJar Transaction Sync” Job
3. Navigate to the Job Steps tab
4. Under Scope, click on the blue “RefArch” button
5. Select “Specific Sites” for the Scope field and then select the sites you would like to sync data to TaxJar from and click Assign.
6. When you are ready to enable the sync, go to the Schedule and History tab and check the Enabled box.

Note: Once enabled with the default configuration from the import, this job will run every ten minutes and sync all orders that have been created or modified since the last sync to TaxJar.

